

THE INITIAL-TO-FINAL-STATE INVERSE PROBLEM WITH TIME-INDEPENDENT POTENTIALS

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ABSTRACT. The initial-to-final-state inverse problem consists in determining a quantum Hamiltonian assuming the knowledge of the state of the system at some fixed time, for every initial state. This problem was formulated by Caro and Ruiz and motivated by the data-driven prediction problem in quantum mechanics. Caro and Ruiz analysed the question of uniqueness for Hamiltonians of the form $-\Delta + V$ with an electric potential $V = V(t, x)$ that depends on the time and space variables. In this context, they proved that uniqueness holds in dimension $n \geq 2$ whenever the potentials are bounded and have super-exponential decay at infinity. Although their result does not seem to be optimal, one would expect at least some degree of exponential decay to be necessary for the potentials. However, in this paper, we show that by restricting the analysis to Hamiltonians with time-independent electric potentials, namely $V = V(x)$, uniqueness can be established for bounded integrable potentials exhibiting only super-linear decay at infinity, in any dimension $n \geq 2$. This surprising improvement is possible because, unlike Caro and Ruiz's approach, our argument avoids the use of complex geometrical optics (CGO). Instead, we rely on the construction of stationary states at different energies—this is possible because the potential does not depend on time. These states will have an explicit leading term, given by a Herglotz wave, plus a correction term that will vanish as the energy grows. Besides the significant relaxation of decay assumptions on the potential, the avoidance of CGO solutions is important in its own right, since such solutions are not readily available in more complicated geometric settings.

1. INTRODUCTION

In quantum mechanics, the family of wave functions $\{u(t, \cdot) : t \in [0, T]\}$ describes the state of the system during an interval of time $[0, T]$. If the motion takes place only under the influence of an electric potential V and the initial state $u(0, \cdot)$ is prescribed by f , then states

$$u : (t, x) \in [0, T] \times \mathbb{R}^n \mapsto u(t, x) \in \mathbb{C}$$

are solutions of the initial-value problem for the Schrödinger equation

$$(1) \quad \begin{cases} i\partial_t u = -\Delta u + V u & \text{in } (0, T) \times \mathbb{R}^n, \\ u(0, \cdot) = f & \text{in } \mathbb{R}^n. \end{cases}$$

It is a well known fact that, if $V \in L^1((0, T); L^\infty(\mathbb{R}^n))$, then for every $f \in L^2(\mathbb{R}^n)$ there exists a unique $u \in C([0, T]; L^2(\mathbb{R}^n))$ solving (1). Moreover, the linear map $f \in L^2(\mathbb{R}^n) \mapsto u \in C([0, T]; L^2(\mathbb{R}^n))$ is bounded. The solutions with the previous properties will be referred to as *physical*.

Using the physical solutions associated to a potential V , Caro and Ruiz formulated in [9] an inverse problem consisting in determining the electric potential from data measured only at the initial and final times. These data were modelled by the *initial-to-final-state map*, which is defined by

$$\mathcal{U}_T : f \in L^2(\mathbb{R}^n) \mapsto u(T, \cdot) \in L^2(\mathbb{R}^n),$$

where u is the solution of the problem (1). This mapping is bounded in $L^2(\mathbb{R}^n)$. In [9], Caro and Ruiz proved that $\mathcal{U}_T$ uniquely determines the potential $V = V(t, x)$ in dimension $n \geq 2$ whenever $V \in L^1((0, T); L^\infty(\mathbb{R}^n))$ has super-exponential decay, that is, $e^{\rho|x|}V \in L^\infty((0, T) \times \mathbb{R}^n)$ for all $\rho > 0$. The reason that explains why the super-exponential decay was needed in [9] is the use of a family of solutions, usually called complex geometrical optics (CGO) solutions, that grow exponentially at infinity. In light

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